

# Non-invasive intracranial pressure estimation from cerebral blood flow dynamics using wavelet-based deep learning

**Speaker:**

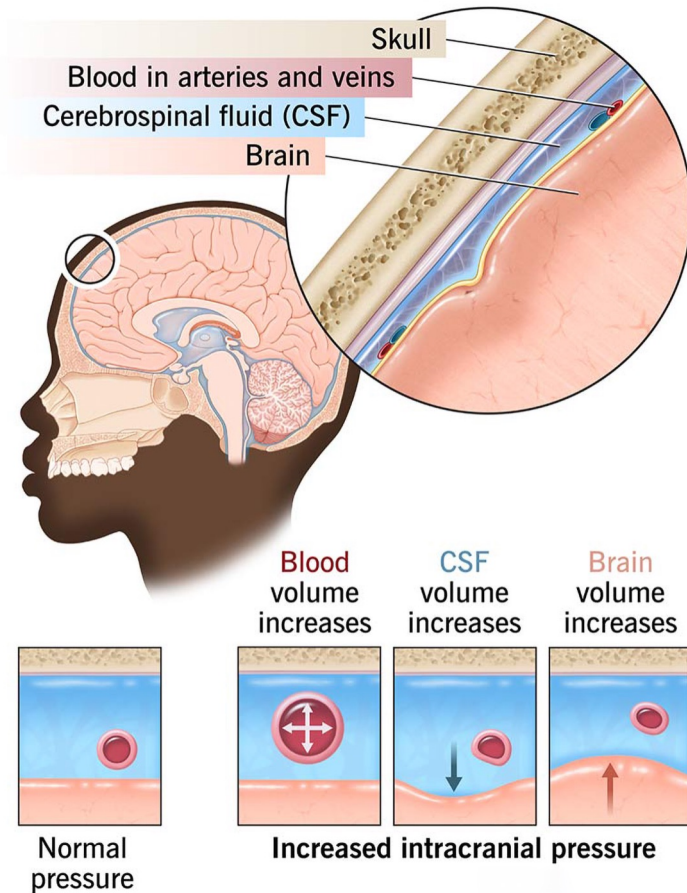


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Diego Fernando Lopez • Marta Peris • Maria Poca • Juan Sahuquillo • Turgut Durduran • Gemma Piella

## INTRO

# WHY BRAIN PRESSURE MATTERS



**Intracranial Pressure (ICP):** The pressure inside the skull, vital for proper brain function.

**Conditions Affected:** Brain trauma, hydrocephalus, craniosynostosis, and other neurological conditions can cause dangerous changes in ICP.

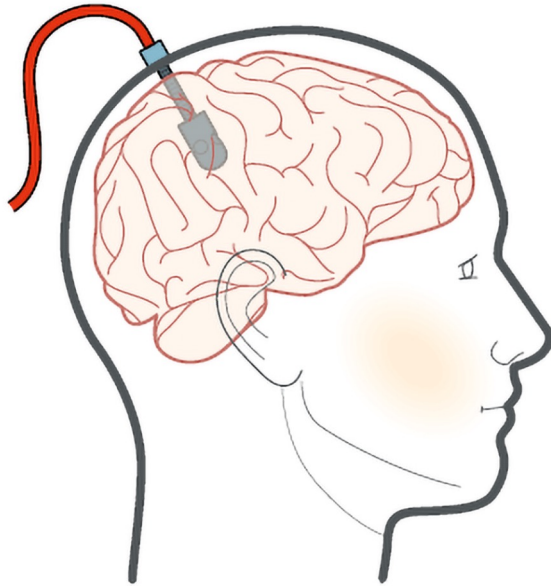
**Impact:** Elevated ICP can damage brain tissue, leading to neurological deficits and even death.

*Source: Cleveland Clinic / Increased Intracranial Pressure (ICP)*

## INTRO

# TODAY'S REALITY • INVASIVE MONITORING

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**Source:** Erasmo Barros da Silva Junior / Brain4Care

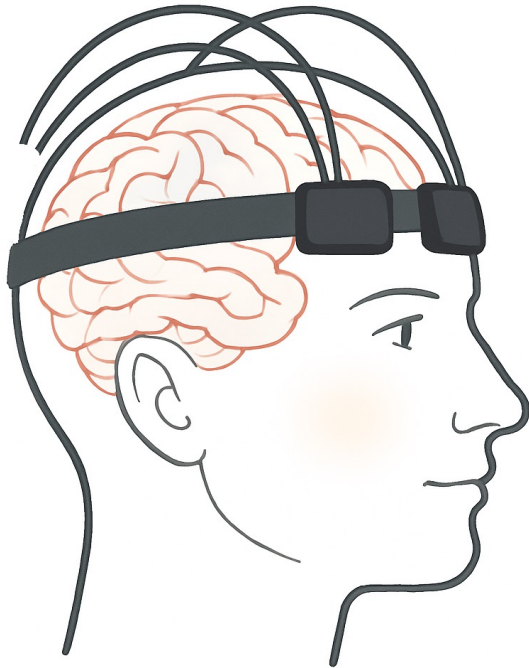
**Current Standard:** Intracranial pressure monitoring typically involves invasive procedures, such as placing a catheter into the brain through a hole drilled in the skull

**Limitations:** These methods carry risks like infection and hemorrhage, require surgical intervention, and are not suitable for all patients

## INTRO

# OUR VISION • NON-INVASIVE MONITORING

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**Innovative Approach:** Leverage non-invasive techniques based on biophotonics and machine learning to monitor intracranial pressure

**Key Features:** Low-cost, easy-to-use, and bedside monitoring without the need for surgical intervention

### **References:**

*Fischer J. et al • Non-invasive ICP estimation using diffuse optics: Proof of concept • 2020*

*Flanders T. M. et al • Optical detection of intracranial pressure and perfusion changes in neonates with hydrocephalus • 2021*

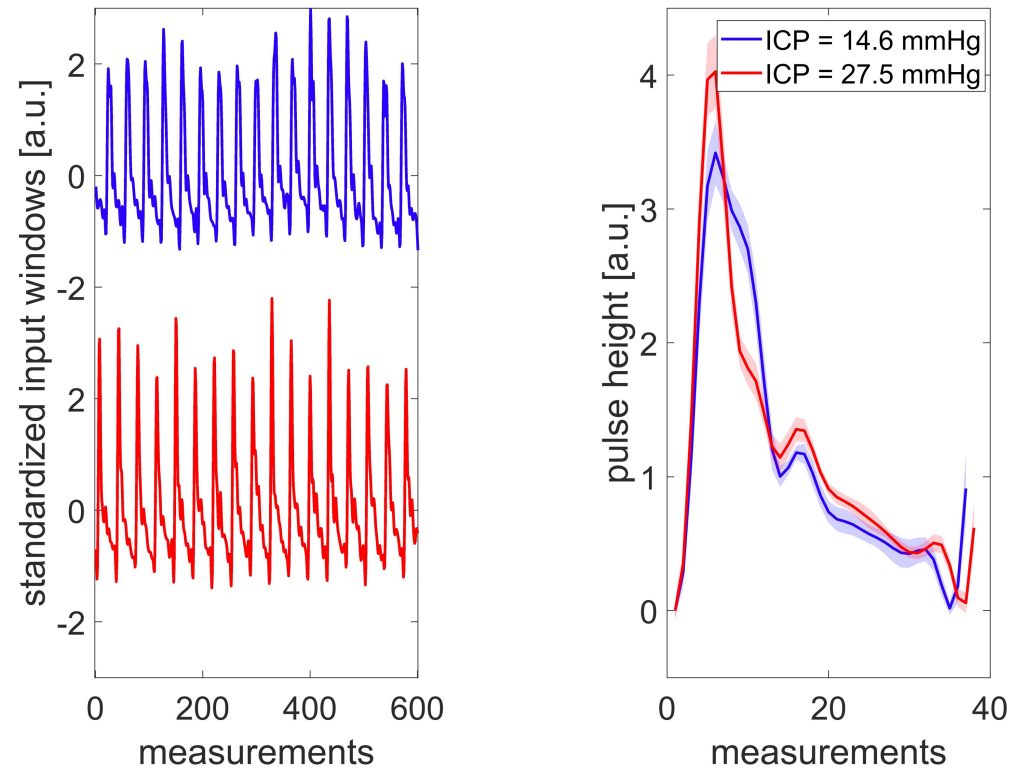
*Wu K. C. et al • Validation of diffuse correlation spectroscopy measures of critical closing pressure against transcranial Doppler ultrasound • 2021*

*Tabassum S. et al • Clinical translation of non-invasive intracranial pressure sensing with diffuse correlation spectroscopy • 2023*

*Baker W. B. et al • Modified Beer-Lambert algorithm to measure pulsatile blood flow, critical closing pressure, and intracranial hypertension • 2024*

## INTRO

# BLOOD FLOW & PRESSURE RELATIONSHIP



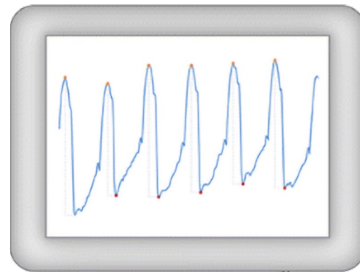
Blood flow time traces in an input window for **low** and **high** intracranial pressure

*The average ICP value corresponding to the window is shown in the legend*

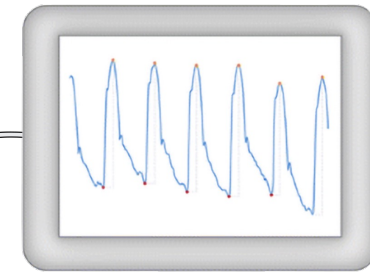
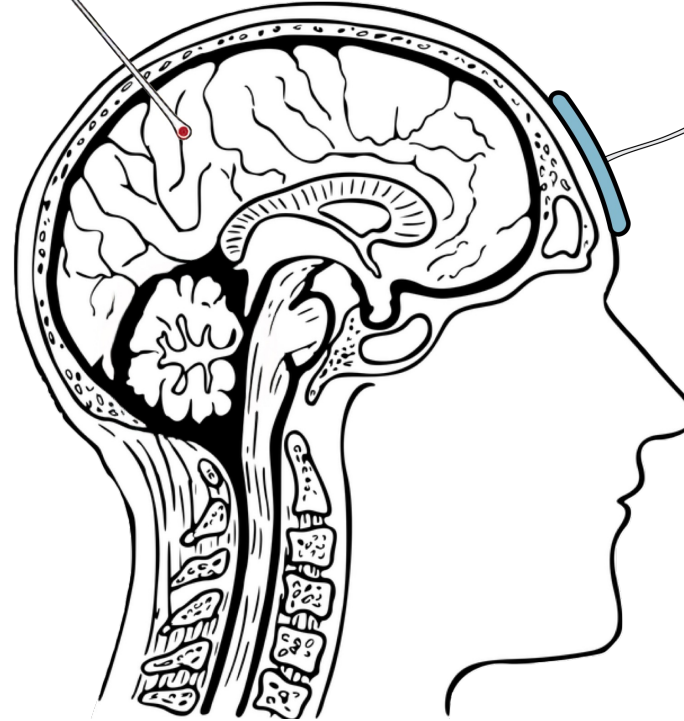
**Reference:** Fischer J. • Transcranial diffuse optical measurements of pulsatility derived parameters for neuromonitoring applications • PhD Thesis • 2021

## INTRO

# BLOOD FLOW → PRESSURE



**Subdural Sensor**  
Ground truth of ICP\*



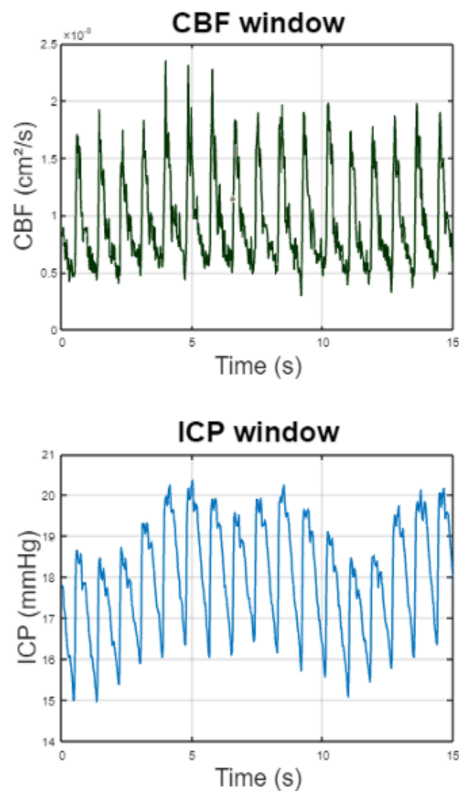
**Non-Invasive Optical Probe**  
CBF\*

\*CBF - cerebral blood flow

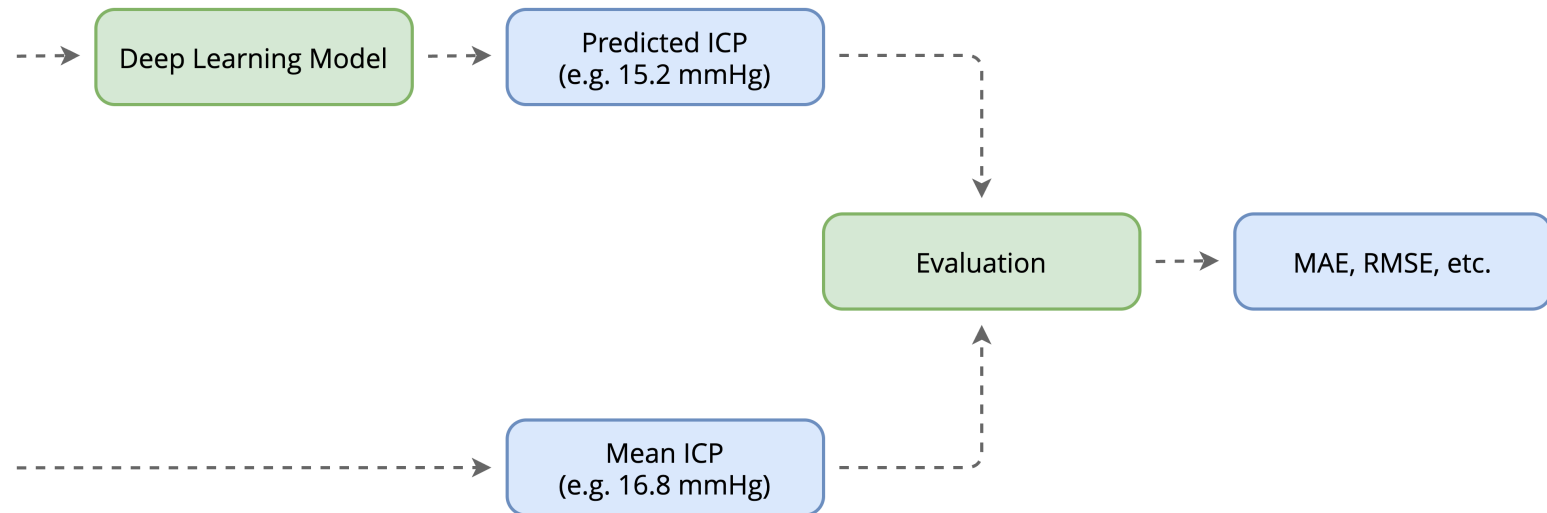
\*ICP - intracranial pressure

# INTRO

## PROPOSED SOLUTION



*Fifteen seconds windows*



*MAE – absolute error*  
*RMSE – root mean square error*

# TEAM CONSORTIUM

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Machine Learning  
Expertise



Biophotonics  
Expertise



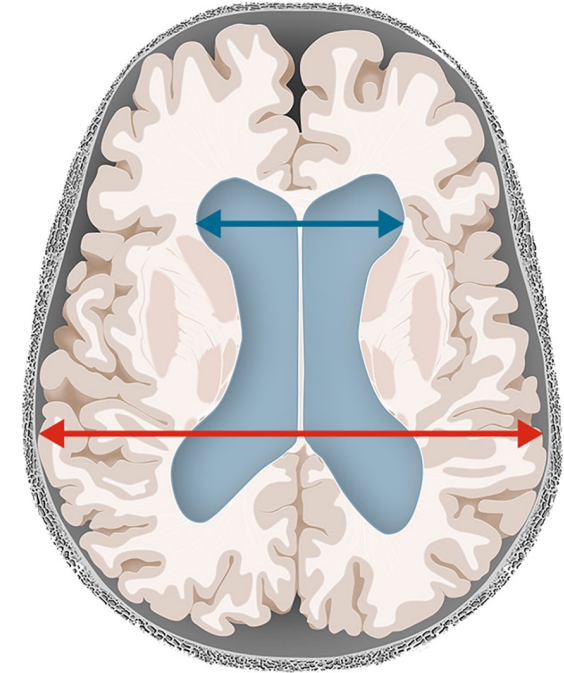
Safe Laser  
Technologies



Clinical Studies and  
Neurosurgery

# IDIOPATHIC NORMAL PRESSURE HYDROCEPHALUS

- **Clinical symptoms:** Gait disturbance, cognitive decline, urinary incontinence
- **Radiological marker:** Evans' Index  $> 0.3$  indicates enlarged ventricles
- **Patient selection:** Indicated for continuous ICP monitoring
- **Monitoring setup:** Subdural sensor used to record long-term ICP waveforms



$$\text{Evans' index} = \frac{\text{maximum width of the frontal horns of the lateral ventricles}}{\text{maximum internal diameter of the skull}}$$

**DATA**  
**PATIENTS**  

---

**22**

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Train + Validation

**22**

---

Test

**DATA**

# **SIGNAL LENGTH**

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**72h**

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Training

**18h**

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Validation

**110h**

---

Test

**DATA**  
**SPLITS**  

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**1**

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Training subset

*Train the model*

**2**

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Validation subset

*Prevent overfitting*

**3**

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Test subset

*Final evaluation*

## MODELS

# HIGH-LEVEL COMPARISON

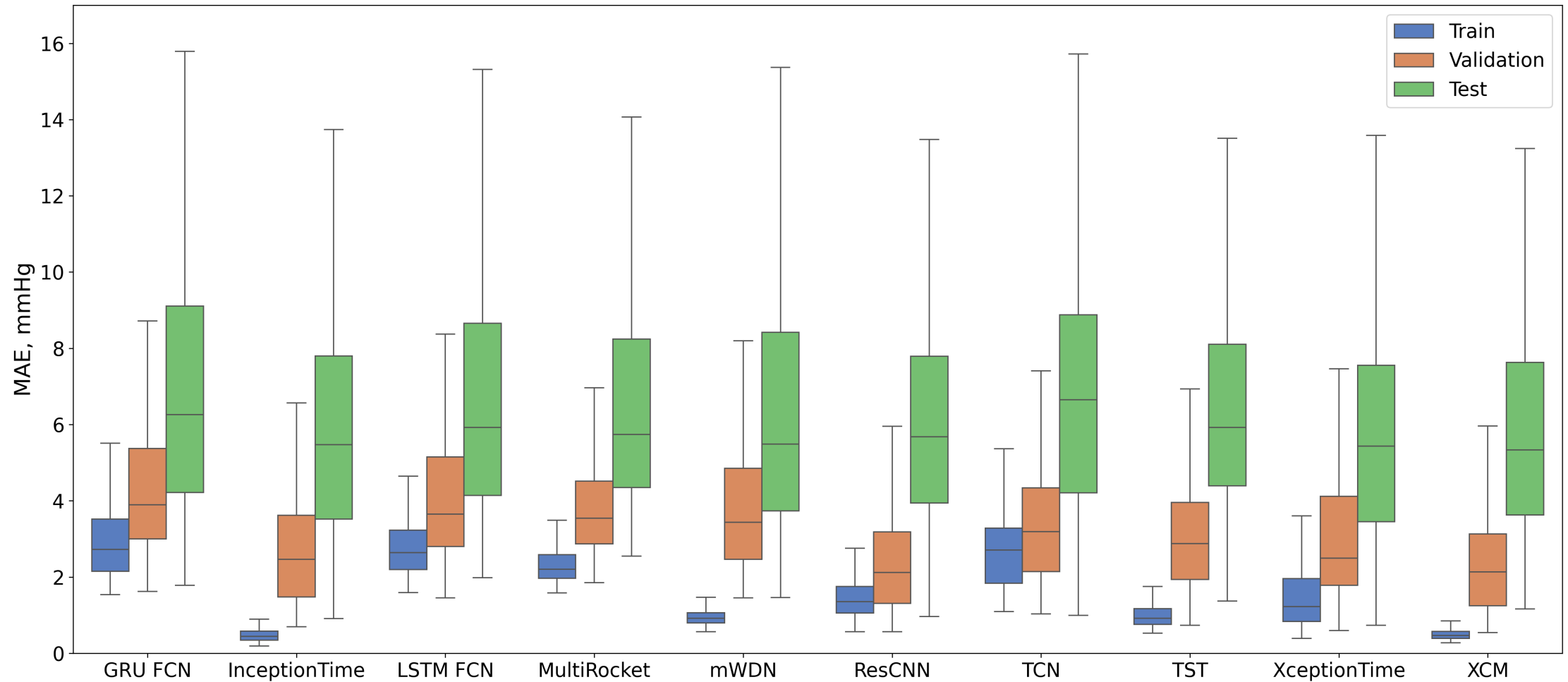
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| #  | Model         | Release Year | Parameters, k | MACs, M |
|----|---------------|--------------|---------------|---------|
| 1  | LSTM-FCN      | 2017         | 785.1         | 318.9   |
| 2  | GRU-FCN       | 2018         | 654.9         | 318.8   |
| 3  | mWDN          | 2018         | 4239.2        | 553.6   |
| 4  | TCN           | 2018         | 66.7          | 187.8   |
| 5  | ResCNN        | 2019         | 256.1         | 309.1   |
| 6  | InceptionTime | 2019         | 388.5         | 470.6   |
| 7  | XceptionTime  | 2019         | 399.1         | 294.1   |
| 8  | TST           | 2021         | 703.4         | 1652.7  |
| 9  | XCM           | 2021         | 615.9         | 740.2   |
| 10 | MultiRocket   | 2022         | 149.2         | 0.1     |

- *Parameters represent the total number of weights within the model, measured in thousands (k)*
- *MAC (Multiply-Accumulate operations) refers to the number of operations required to perform the matrix multiplications in a model, measured in millions (M)*

# RESULTS

## MODEL PERFORMANCE



## RESULTS

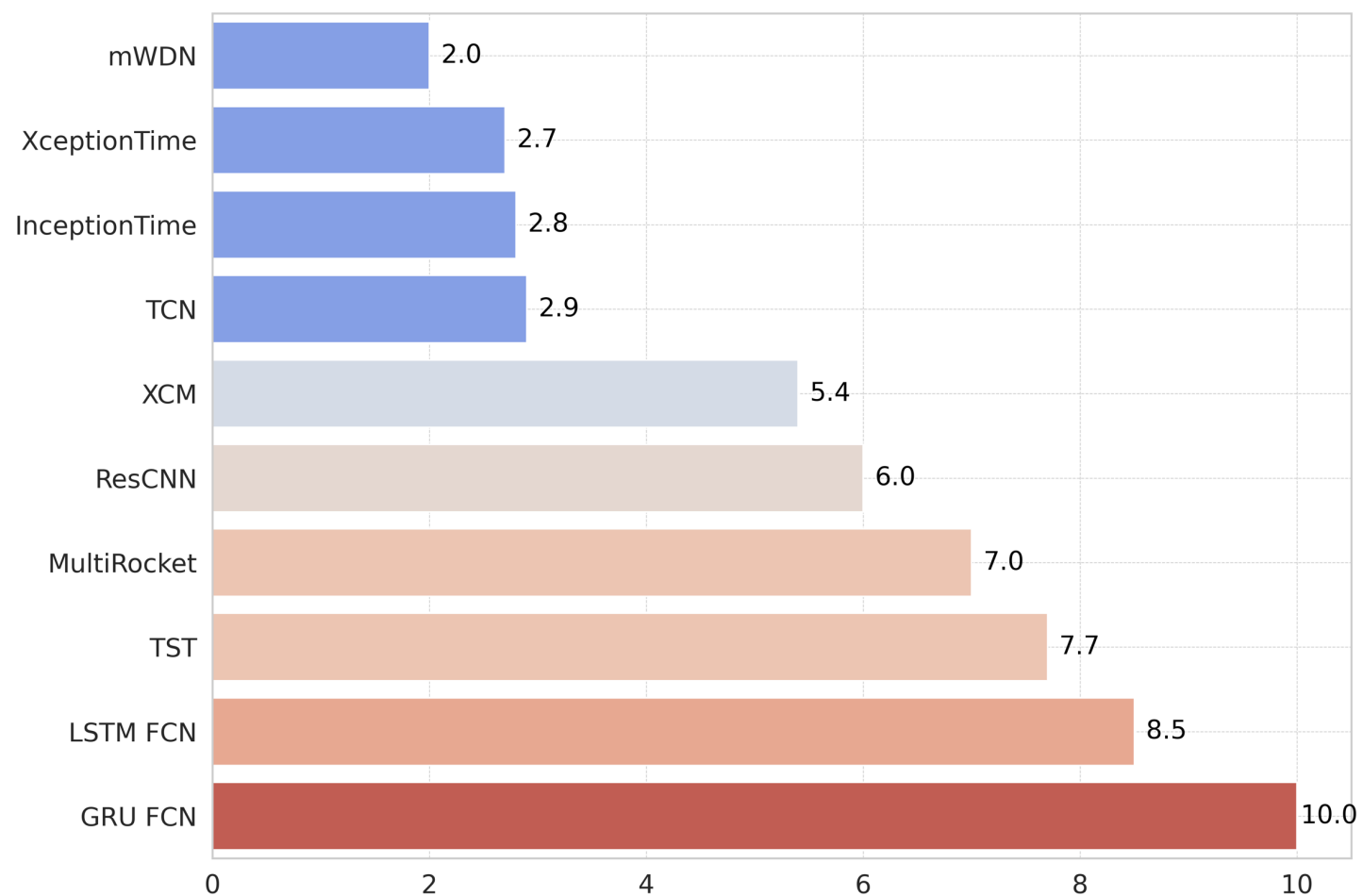
# WEIGHTED RANKING SCORES

$$\text{Model Rank} = \frac{w_{train} \times r_{train} + w_{val} \times r_{val} + w_{test} \times r_{test}}{w_{train} + w_{val} + w_{test}}$$

$$w_{train} = 0.1$$

$$w_{val} = 0.4$$

$$w_{test} = 0.5$$



## RESULTS

### **BEST MODEL • mWDN\***

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**0.5 mmHg**

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Training

**2.6 mmHg**

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Validation

**5.8 mmHg**

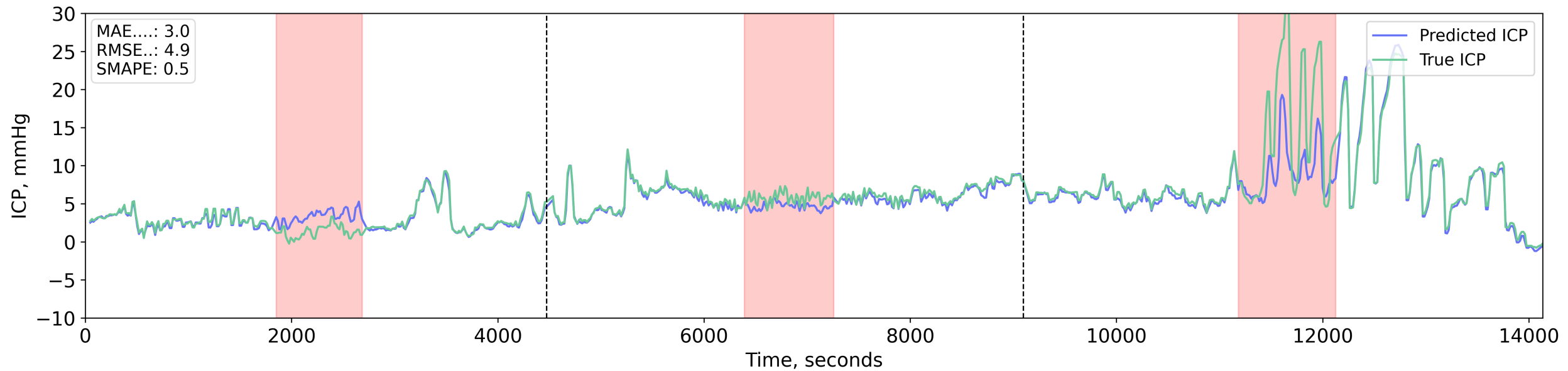
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Test

\*mWDN - multilevel Wavelet Decomposition Network

# RESULTS

## VALIDATION • SUBJECT 17



**MAE:** mean absolute error

**RMSE:** root mean square error

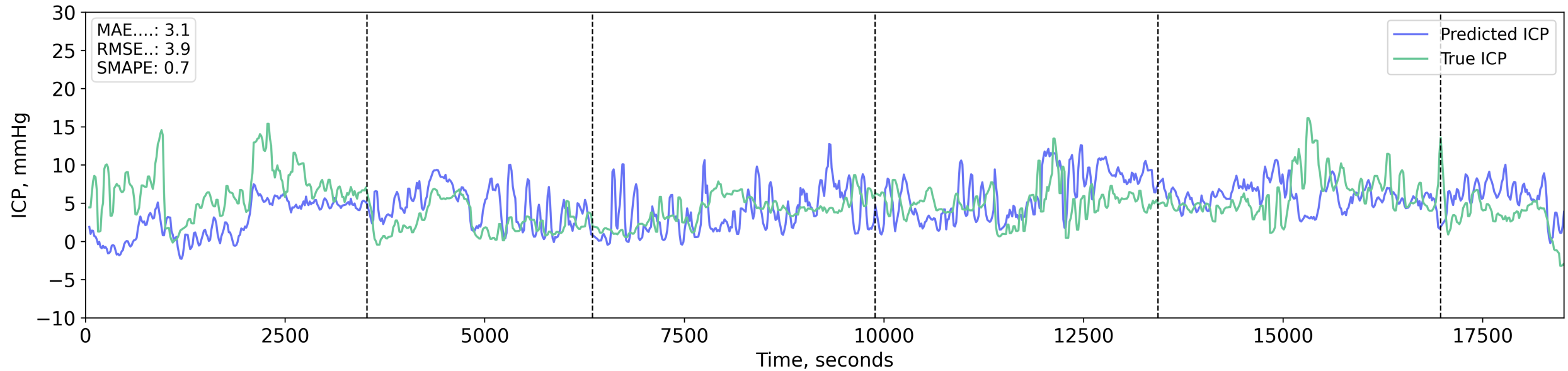
**SMAPE:** symmetric absolute percentage error

Training

Validation

# RESULTS

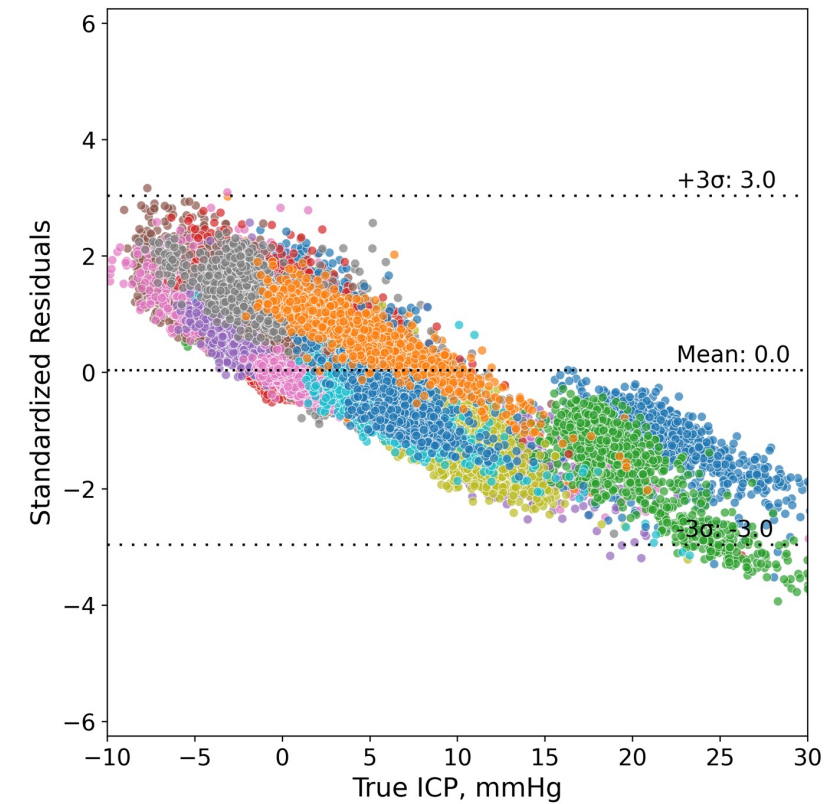
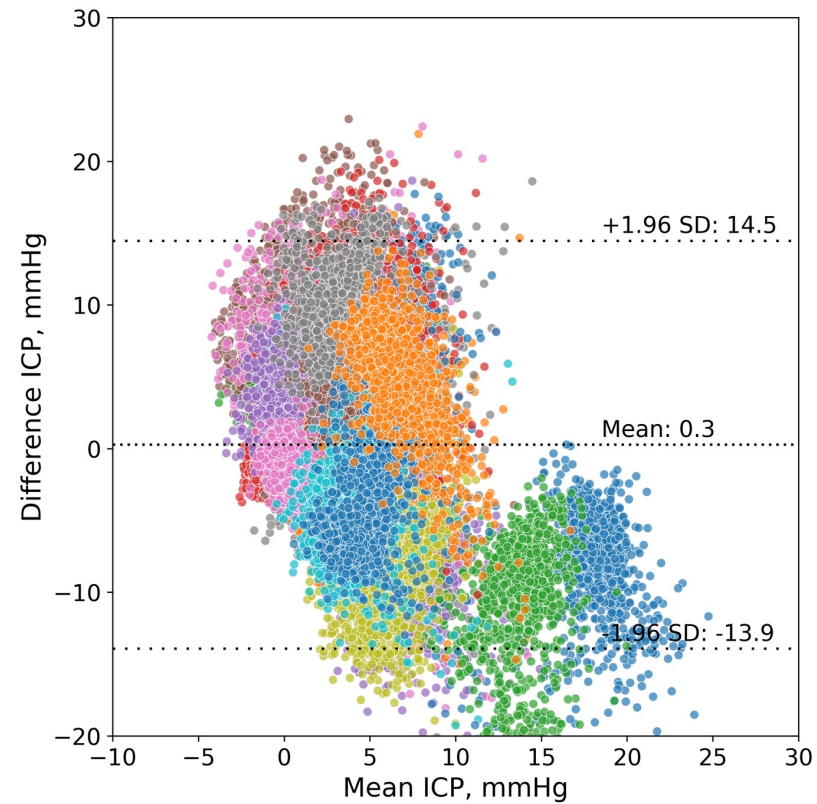
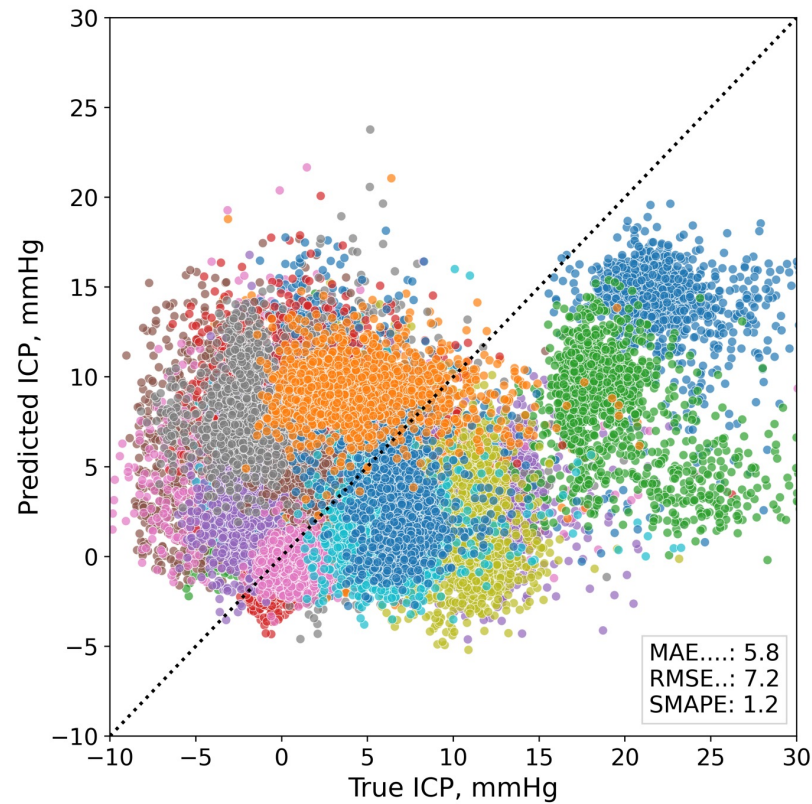
## TEST • SUBJECT 42



**MAE:** mean absolute error  
**RMSE:** root mean square error  
**SMAPE:** symmetric absolute percentage error

# RESULTS

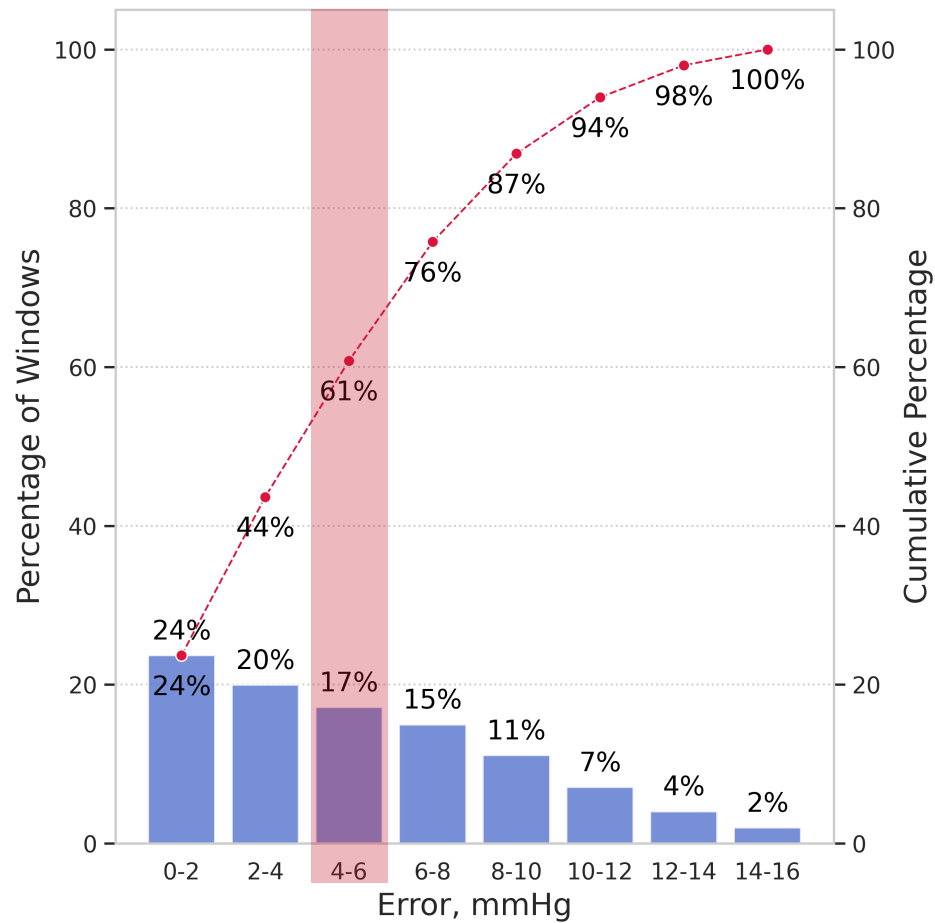
## ALL TEST SUBJECTS



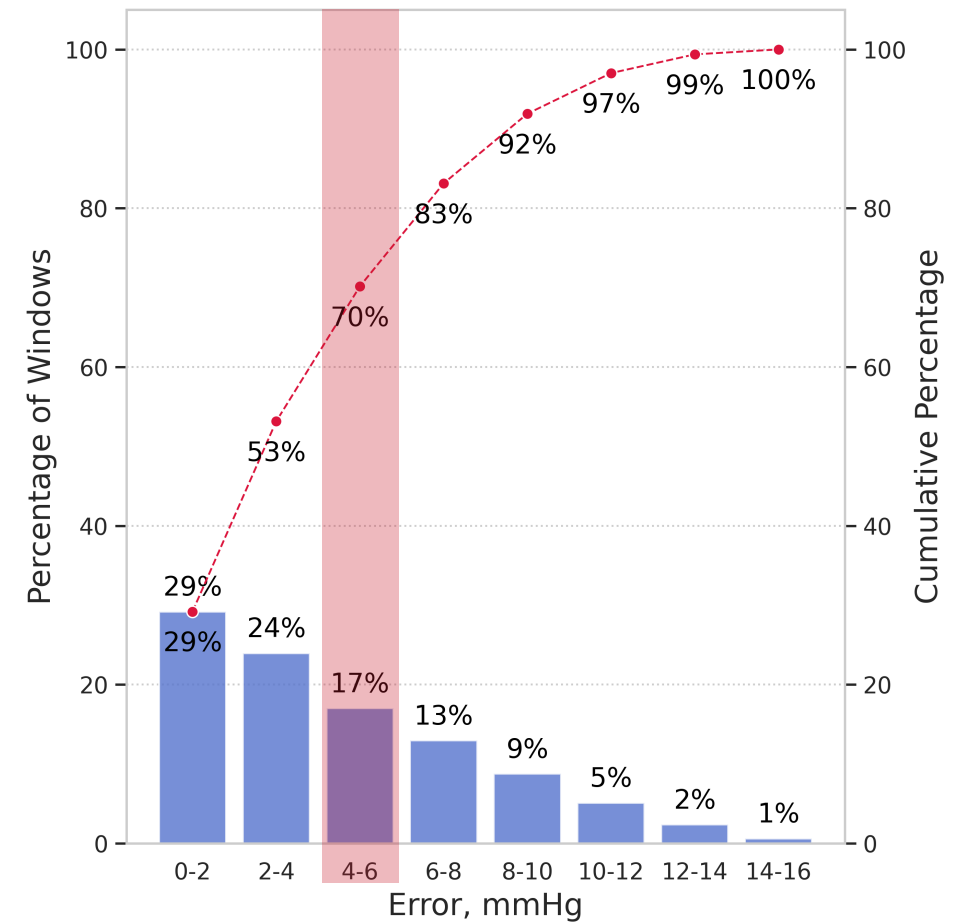
*Each color represents one subject*

# RESULTS

## ERROR DISTRIBUTION



Full ICP range: from -10 to 30 mmHg



Truncated ICP range: from 0 to 15 mmHg

## RESULTS

# STANDARDS VS PERFORMANCE

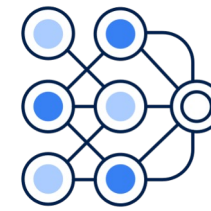
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### STANDARDS\*

0–20 mmHg:  $\pm 2$  mmHg

>20 mmHg:  $\pm 10\%$



### ERROR

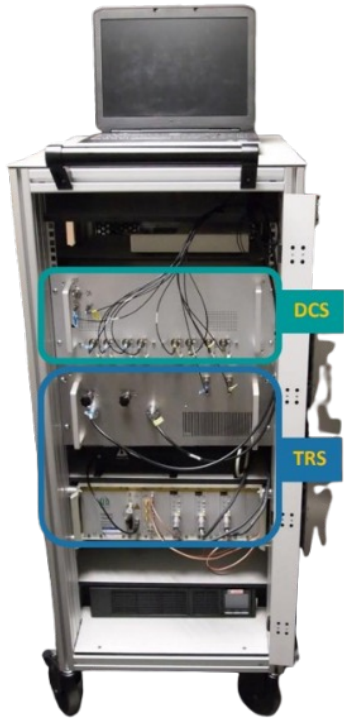
Validation: 2.6 mmHg

Test: 5.8 mmHg

*\*According to the **American National Standards Institute** and the **Association for the Advancement of Medical Instrumentation***

# CONCLUSION

## FUTURE WORK



**Current device**

DCS + TRS cart



**New device**

SCOS unit



### New device deployment

- Speckle contrast optical spectroscopy (SCOS)
- Safe, low cost and high signal-to-noise ratio



### Hospital collaborations

- Expand data collection
- Multi-center study scale-up



### Methodological improvements

- Data selection
- Artifact removal
- Model architecture

\* DCS - Diffuse Correlation Spectroscopy

\* TRS - Time resolved spectroscopy

## CONCLUSION

# EXPANDING DATA COLLECTION

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| # | Cohort                               | Measurement Type | Target | Status                                                                                         |
|---|--------------------------------------|------------------|--------|------------------------------------------------------------------------------------------------|
| 1 | Acute brain injuries (TBI + stroke*) | ICP + CBF        | 70     |  Recruiting |
| 2 | Idiopathic NPH                       | ICP + CBF        | 80     |  Recruiting |
| 3 | Pediatric craniosynostosis           | CBF              | 40     |  Recruiting |
| 4 | Pediatric hydrocephalus              | ICP + CBF        | 40     |  Recruiting |
| 5 | Healthy controls                     | CBF              | 30     |  Planned    |

*\* For stroke-affected patients, the only CBF is collected*

## CONCLUSION

# RELATED TALKS

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**Mirko Fornasier, MSc**

Advances in Instrumentation and Technology

**M1B.5.** A compact, self-contained module for fast and non-invasive blood flow monitoring



**Lisa Kobayashi, PhD**

Theory, Algorithms, and Modeling

**Th3B.3.** Algorithm for deriving absolute blood flow from reduced speckle contrast data

# Let's get in touch



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